

Ordinary Differential Equations

Master Cheat Sheet & Review Guide

Chapters 1–3 • First-Order Methods, Qualitative Analysis & Second-Order Linear Theory

Classification & Vocabulary • Separation of Variables (Growth/Decay & Newton's Law of Cooling)
First-Order Linear (Integrating Factor) • Exact Equations • Slope Fields
Autonomous Equations & Phase Lines • Euler's Method • Existence & Uniqueness
Linear Independence & the Wronskian • Characteristic Equation (Three Root Cases)
Undetermined Coefficients • Mechanical Vibrations

How to read this guide. Explanatory prose is set in ordinary black text. Every *live* mathematical step — each calculation, substitution, and algebraic manipulation you would write out by hand — is set in **purple ink**, exactly as a professor would mark worked solutions on the board. When you see purple, you are watching mathematics being *done*, not described. Boxed formulas carry an **orange** accent; section headers and theorems are in **blue**.

This document has three parts. **Part I** is a condensed cheat sheet — every formula, test, and method on a single sweep. **Part II** is a graded review sequence of practice questions covering everything in the notes. **Part III** is a complete answer key with fully worked, step-by-step solutions.

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A first-principles companion to the Chapters 1–3 whiteboard notes

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Part I | Condensed Cheat Sheet

One pass through every definition, test, formula, and method.

1 Vocabulary & Classification

Order = order of the *highest* derivative present. **Degree** = power to which the highest-order derivative is raised, *after* clearing radicals/fractions (only defined when the equation is polynomial in its derivatives).

ODE vs. PDE. An ODE involves derivatives with respect to *one* independent variable; a PDE involves partials in two or more, e.g. $\partial f / \partial x$, $\partial f / \partial y$.

Linear n th-order ODE — can be written

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = g(x).$$

Linearity requires: (i) y and each derivative appear to the **first power only**; (ii) **no products** of y with its derivatives; (iii) **no transcendental functions** of y ($\sin y$, e^y , $\ln y$, $\dots$). Coefficients $a_k(x)$ and forcing $g(x)$ may be *any* functions of x .

Common Pitfall

$\sqrt{x+y} \neq \sqrt{x} + \sqrt{y}$. A radical or square containing y or y' is nonlinear — always rationalize first, then read off order and degree.

2 Solutions, IVPs & Verification

General solution carries arbitrary constants (n of them for an n th-order ODE); a **particular solution** fixes them. An **initial value problem (IVP)** supplies data $y(x_0), y'(x_0), \dots$ to pin the constants.

Verification by substitution. To confirm $y = \varphi(x)$ solves an ODE, compute the needed derivatives, substitute, and check both sides agree identically. (e.g. $y = e^{-2x} \sin 3x$ satisfies $y'' + 4y' + 13y = 0$.)

3 Separation of Variables

If $\frac{dy}{dx} = g(x)h(y)$, then $\int \frac{dy}{h(y)} = \int g(x) dx + C$.

Exponential growth & decay. $\frac{dy}{dt} = ky \Rightarrow y = y_0 e^{kt}$. $k > 0$ growth, $k < 0$ decay. Half-life $t_{1/2} = \frac{\ln 2}{|k|}$; doubling time $\frac{\ln 2}{k}$.

Population / momentum / radioactivity. $\frac{dP}{dt} = rP \Rightarrow P = Ke^{rt}$; $\frac{dN}{dt} = -\lambda N \Rightarrow N = N_0 e^{-\lambda t} \rightarrow 0$.

Key Idea

Newton's Law of Cooling. The rate of temperature change is proportional to the gap with the ambient temperature T_m :

$$\frac{dT}{dt} = -k(T - T_m), \quad k > 0.$$

Separating: $\int \frac{dT}{T - T_m} = -k \int dt \Rightarrow \ln |T - T_m| = -kt + C$, hence

$$T(t) = T_m + (T_0 - T_m) e^{-kt}$$

As $t \rightarrow \infty$, $T \rightarrow T_m$. The same template models warming (object colder than room).

4 First-Order Linear ODEs (Integrating Factor)

Standard form: $y' + P(x)y = Q(x)$. **Integrating factor:** $\mu(x) = e^{\int P(x) dx}$.

$$(\mu y)' = \mu Q \implies y = \frac{1}{\mu} \left[\int \mu Q(x) dx + C \right].$$

The trick rests on the product rule: $\mu y' + \mu P y = (\mu y)'$ exactly when $\mu' = \mu P$, i.e. $\mu = e^{\int P}$.

Step-by-Step Methodology

1. Check it is first-order linear; if separable, that is usually easier.
2. Put in standard form $y' + P(x)y = Q(x)$ (*divide by the leading coefficient*).
3. Compute $\mu = e^{\int P dx}$.
4. Write $y = \frac{1}{\mu} [\int \mu Q dx + C]$ and integrate.
5. Apply any initial condition to solve for C .

5 Exact Equations

Write the equation as $M(x, y) dx + N(x, y) dy = 0$. It is the total differential of a level curve $f(x, y) = C$, since $df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 0$.

Exactness test: $M dx + N dy = 0$ is exact $\iff \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ (because $f_{xy} = f_{yx}$).
Then $M = f_x$, $N = f_y$, and the solution is $f(x, y) = C$.

Step-by-Step Methodology

1. Identify M, N ; verify $M_y = N_x$.
2. Integrate $f = \int M dx = (\text{terms}) + g(y)$.
3. Differentiate w.r.t. y , set $f_y = N$ to solve for $g'(y)$, integrate for $g(y)$.
4. State the implicit solution $f(x, y) = C$.

6 Slope Fields & Qualitative Analysis

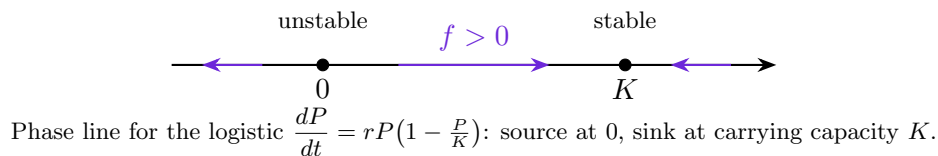
A first-order ODE $\frac{dy}{dx} = f(x, y)$ assigns a slope to each point of the plane. **Isoclines** are curves $f(x, y) = \text{const}$ (constant slope); the **nullcline** / **equilibrium** slope is where $f = 0$ (horizontal segments). Trajectories cannot cross. Reading long-run behavior (horizontal asymptotes, the *range* of a solution curve as $x \rightarrow \infty$) is done straight off the field.

7 Autonomous Equations & Phase Lines

Autonomous: $\frac{dy}{dt} = f(y)$ — no explicit t . **Equilibria** solve $f(y^*) = 0$.

Stability (linearization on the line): $f'(y^*) < 0 \Rightarrow$ **stable** (sink); $f'(y^*) > 0 \Rightarrow$ **unstable** (source); $f'(y^*) = 0 \Rightarrow$ inconclusive (check signs — may be semi-stable).

The **phase line** marks equilibria and the sign of f between them (arrows up where $f > 0$, down where $f < 0$).

**8 Euler's Method (Numerical)**

Step size $h = \Delta t$. From the tangent line $y - y_n = f(t_n, y_n)(t - t_n)$:

$$y_{n+1} = y_n + h f(t_n, y_n), \quad t_{n+1} = t_n + h.$$

Local truncation error is $O(h^2)$; global error $O(h)$ — halving h roughly halves total error. Smaller steps $\Rightarrow$ more accurate (and closer to the exact value, e.g. $B(1) = 1300e^{0.06}$ for $B' = 0.06B$).

9 Existence & Uniqueness**Theorem: First-Order (Picard–Lindelöf)**

For $y' = f(t, y)$, $y(t_0) = y_0$: if f and $\partial f / \partial y$ are continuous on a rectangle containing (t_0, y_0) , then a *unique* solution exists on some interval about t_0 .

Theorem: First-Order Linear

For $y' + p(t)y = g(t)$, $y(t_0) = y_0$: if p, g are continuous on an open interval $I \ni t_0$, a unique solution exists on *all* of I .

Theorem: Second-Order Linear

For $y'' + p(t)y' + q(t)y = g(t)$ with $y(t_0) = b_0$, $y'(t_0) = b_1$: if p, q, g are continuous on $I \ni t_0$, a unique solution exists throughout I .

Common Pitfall

$y' = y^{1/3}$, $y(0) = 0$ has f continuous but $\partial f / \partial y = \frac{1}{3}y^{-2/3}$ *discontinuous* at $y = 0$ — uniqueness fails: both $y \equiv 0$ and $y = (\frac{2}{3}t)^{3/2}$ solve it.

10 Linear Independence & the Wronskian

Linearly independent means neither function is a scalar multiple of the other (e.g. x^2 & x are independent; $2x$ & x are dependent).

Wronskian: $W(y_1, y_2)(t) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_1' y_2$.

If y_1, y_2 solve the same linear ODE and $W(t_0) \neq 0$ for some t_0 , they form a **fundamental set** (independent); the general solution is $y = c_1 y_1 + c_2 y_2$.

Theorem: Abel's Formula

For $y'' + p(t)y' + q(t)y = 0$, $W(t) = W(t_0) e^{-\int_{t_0}^t p(s) ds}$. Hence W is *either identically zero or never zero* on I .

11 Second-Order Linear, Constant Coefficients

Superposition. If y_1, y_2 solve $y'' + py' + qy = 0$, so does $c_1 y_1 + c_2 y_2$.

Method. Guess $y = e^{rt}$ in $ay'' + by' + cy = 0$; since $e^{rt} \neq 0$, the bracket gives the **characteristic equation**

$$ar^2 + br + c = 0.$$

Roots of $ar^2 + br + c = 0$	General solution $y(t)$
Distinct real $r_1 \neq r_2$	$c_1 e^{r_1 t} + c_2 e^{r_2 t}$
Repeated real r	$(c_1 + c_2 t) e^{rt}$
Complex $r = \alpha \pm \beta i$	$e^{\alpha t} (c_1 \cos \beta t + c_2 \sin \beta t)$

The repeated-root factor t and the complex case both come from one source; complex roots use **Euler's formula** $e^{i\theta} = \cos \theta + i \sin \theta$.

12 Undetermined Coefficients (Non-homogeneous)

General solution $y = y_c + y_p$: complementary y_c solves the homogeneous part; particular y_p is *guessed* to match the forcing $g(t)$.

Forcing $g(t)$	Trial y_p
$K e^{at}$	$A e^{at}$
degree- n polynomial	$A_n t^n + \cdots + A_1 t + A_0$
$K \sin bt$ or $K \cos bt$	$A \cos bt + B \sin bt$
products of the above	corresponding product form

Common Pitfall

Overlap / resonance. If the trial y_p already appears in y_c , multiply the guess by t (or t^2 for a double root). Example: for $y'' - 3y' + 2y = e^t$ with $y_c = c_1 e^t + c_2 e^{2t}$, use $y_p = A t e^t$.

13 Mechanical Vibrations

Spring–mass–damper: $m u'' + \gamma u' + k u = F(t)$.

Hooke at equilibrium: $mg = kL$ (mass stretches spring by length L). $\Rightarrow k = \frac{mg}{L}$.

Free undamped ($\gamma = 0, F = 0$): $u'' + \omega_0^2 u = 0$, $\omega_0 = \sqrt{k/m}$,

$$u = c_1 \cos \omega_0 t + c_2 \sin \omega_0 t = R \cos(\omega_0 t - \delta), \quad R = \sqrt{c_1^2 + c_2^2}.$$

Free damped: roots of $mr^2 + \gamma r + k = 0$; discriminant $\gamma^2 - 4mk$ decides the regime:

Condition	Roots	Regime
$\gamma^2 > 4mk$	distinct real, < 0	overdamped (no oscillation)
$\gamma^2 = 4mk$	repeated real, < 0	critically damped
$\gamma^2 < 4mk$	complex, $\text{Re} < 0$	underdamped (decaying oscillation)

Underdamped solution $u = R e^{-\gamma t/2m} \cos(\mu t - \delta)$ with **quasi-frequency** $\mu = \frac{\sqrt{4mk - \gamma^2}}{2m}$.

Forced & resonance. A periodic $F(t)$ drives the system; amplitude peaks when the forcing frequency nears ω_0 (**resonance**; unbounded if undamped). Close-but-unequal frequencies produce **beats**.

Part II | Review Guide: Practice Questions

Work each on a separate sheet; full justification expected. Solutions in Part III.

14 Classification

State the *order*, *degree*, and *linearity* of each.

1. $(y'')^3 + (y')^5 + y = 0$
2. $3x^2y''' + e^xy'' + 2y' + 8 = 0$
3. $y'' + \sin y = 0$
4. $\sqrt{1 + (y')^2} = y''$ (Hint: rationalize first.)

15 Verification

5. Verify by direct substitution that $y = e^{-2x} \sin(3x)$ is a solution of $y'' + 4y' + 13y = 0$.

16 Separation of Variables

6. Solve the IVP $\frac{dP}{dt} = 0.5P$, $P(0) = 200$.
7. (**Newton's Law of Cooling.**) Coffee at 95°C sits in a 20°C room and cools to 80°C after 5 minutes. Find $T(t)$ and the time to reach 60°C .
8. (**Decay.**) Carbon-14 has half-life 5730 yr. Find the decay constant λ and the fraction remaining after 2000 yr.

17 First-Order Linear (Integrating Factor)

9. Solve $\frac{dy}{dx} - 4y = e^{5x}$.
10. Solve $t^2y' + 2ty = \sin t$.
11. Solve the IVP $xy' + 2y = 4x^2$, $y(1) = 2$.

18 Exact Equations

12. Solve $2xy\,dx + (x^2 + 3y^2)\,dy = 0$.
13. Solve $(\cos x - y \sin x)\,dx + \cos x\,dy = 0$.

19 Slope Fields & Qualitative Analysis

14. A slope field has *horizontal* segments exactly along the vertical line $x = -1$. Which equation produces this: $\frac{dy}{dx} = \frac{x}{y}$, $\frac{x-1}{y}$, $\frac{x+1}{y}$? Justify, and state where the field is vertical.
15. For $\frac{dy}{dx} = y(2-y)$, describe the long-run behavior of solutions starting at $y(0) = 0.5$, $y(0) = 3$, and $y(0) = 2$.

20 Autonomous Equations & Phase Lines

16. For $\frac{dy}{dt} = y^2 - 4$: find all equilibria, classify each by the sign of f' , and draw the phase line.
17. For the logistic model $\frac{dP}{dt} = 0.5P\left(1 - \frac{P}{100}\right)$: find the equilibria, classify them, and state the population of fastest growth.

21 Euler's Method

18. For $B'(t) = 0.06B$, $B(0) = 1300$, estimate $B(1)$ using Euler with $\Delta t = 0.5$ (two steps). Show the recursive multiplier.
19. Repeat with a single step $\Delta t = 1$, then compare both estimates to the exact value $B(1) = 1300e^{0.06}$.

22 Existence & Uniqueness

20. Does $y' = y^{1/3}$, $y(0) = 0$ have a unique solution? Explain using the theorem, and exhibit the failure if any.
21. For $(t - 3)y' + (\ln t)y = 2t$, $y(1) = 5$, find the largest interval on which a unique solution is guaranteed.

23 Linear Independence & the Wronskian

22. Compute $W(e^{3t}, e^{-2t})$ and state whether the pair is independent.
23. Are $y_1 = t$ and $y_2 = 3t$ linearly independent? Use the Wronskian and confirm with the definition.
24. Use Abel's formula to find W (up to a constant) for $t^2y'' - 3ty' + 4y = 0$, and check it against the known solution pair $y_1 = t^2$, $y_2 = t^2 \ln t$.

24 Second-Order Homogeneous (Three Cases)

25. Solve $y'' + 8y' + 15y = 0$.
26. Solve $y'' + 6y' + 9y = 0$.
27. Solve $y'' + y' + y = 0$.
28. Solve the IVP $y'' - 4y = 0$, $y(0) = 1$, $y'(0) = 2$.
29. Solve the IVP $y'' - y = 0$, $y(0) = 2$, $y'(0) = 0$.

25 Undetermined Coefficients

30. Solve $y'' - 3y' + 2y = 4e^{5t}$.
31. Find the general solution of $y'' + 4y = 3t^2 - 1$.
32. State (do not solve) the correct trial form y_p for $y'' - 3y' + 2y = e^t$, and explain the modification needed.

26 Mechanical Vibrations

33. A 2-kg mass stretches a spring 0.5 m ($g = 9.8$). Find k , the natural frequency ω_0 , and the period of free undamped motion.
34. For $u'' + \gamma u' + 4u = 0$ ($m = 1, k = 4$), classify the damping for $\gamma = 5, 4, 2$; for the oscillatory case give the quasi-frequency.
35. Write the general solution of $u'' + 2u' + 2u = 0$ and identify the regime and quasi-frequency.

Part III | Answer Key & Worked Solutions

Every live step in purple, exactly as worked by hand.

27 Classification

1. $(y'')^3 + (y')^5 + y = 0$: highest derivative is $y'' \Rightarrow$ **order 2**; it is raised to the third power $\Rightarrow$ **degree 3**; a derivative raised to a power $\Rightarrow$ **nonlinear**.

2. $3x^2y''' + e^xy'' + 2y' + 8 = 0$: highest is $y''' \Rightarrow$ **order 3**, appears to first power $\Rightarrow$ **degree 1**. Each derivative is first-power, no products, no transcendental functions *of* y ; variable coefficients in x are allowed $\Rightarrow$ **linear** (nonhomogeneous).

3. $y'' + \sin y = 0$: **order 2, degree 1**, but $\sin y$ is transcendental in $y \Rightarrow$ **nonlinear**.

4. Rationalize $\sqrt{1 + (y')^2} = y''$:

$$1 + (y')^2 = (y'')^2.$$

Highest derivative y'' to the second power $\Rightarrow$ **order 2, degree 2, nonlinear**.

28 Verification

5. With $y = e^{-2x} \sin 3x$:

$$\begin{aligned} y' &= e^{-2x}(-2 \sin 3x + 3 \cos 3x), \\ y'' &= e^{-2x}(-5 \sin 3x - 12 \cos 3x). \end{aligned}$$

Substitute:

$$\begin{aligned} y'' + 4y' + 13y &= e^{-2x} [(-5 \sin 3x - 12 \cos 3x) + 4(-2 \sin 3x + 3 \cos 3x) + 13 \sin 3x] \\ &= e^{-2x} [(-5 - 8 + 13) \sin 3x + (-12 + 12) \cos 3x] = 0. \checkmark \end{aligned}$$

29 Separation of Variables

6. $\frac{dP}{dt} = 0.5P$ is exponential growth:

$$P(t) = P_0 e^{0.5t} = 200 e^{0.5t}.$$

7. Newton's cooling with $T_m = 20$, $T_0 = 95$:

$$T(t) = 20 + (95 - 20)e^{-kt} = 20 + 75e^{-kt}.$$

Use $T(5) = 80$:

$$\begin{aligned} 80 &= 20 + 75e^{-5k} \Rightarrow 60 = 75e^{-5k} \Rightarrow e^{-5k} = \frac{60}{75} = 0.8, \\ k &= -\frac{1}{5} \ln 0.8 = \frac{1}{5} \ln \frac{5}{4} \approx 0.0446 \text{ min}^{-1}. \end{aligned}$$

Time to 60° :

$$\begin{aligned} 60 &= 20 + 75e^{-kt} \Rightarrow e^{-kt} = \frac{40}{75} = \frac{8}{15}, \\ t &= \frac{\ln(15/8)}{k} = \frac{5 \ln(15/8)}{\ln(5/4)} \approx \frac{0.6286}{0.0446} \approx 14.1 \text{ min.} \end{aligned}$$

8. Decay constant from half-life $t_{1/2} = 5730$:

$$\lambda = \frac{\ln 2}{5730} \approx 1.21 \times 10^{-4} \text{ yr}^{-1}.$$

Fraction after 2000 yr:

$$\frac{N}{N_0} = e^{-\lambda(2000)} = e^{-0.2419} \approx 0.785 \text{ (about 78.5\%)}.$$

30 First-Order Linear (Integrating Factor)

9. $y' - 4y = e^{5x}$: $P = -4$, $\mu = e^{-4x}$.

$$\begin{aligned} (\mu y)' &= \mu e^{5x} = e^x \Rightarrow \mu y = \int e^x dx = e^x + C, \\ y &= e^{4x}(e^x + C) = e^{5x} + Ce^{4x}. \end{aligned}$$

10. $t^2 y' + 2ty = \sin t$: the left side is already $(t^2 y)'$ (product rule).

$$\begin{aligned} (t^2 y)' &= \sin t \Rightarrow t^2 y = -\cos t + C, \\ y &= \frac{-\cos t + C}{t^2}. \end{aligned}$$

11. $xy' + 2y = 4x^2 \Rightarrow y' + \frac{2}{x}y = 4x$. $\mu = e^{\int 2/x dx} = e^{2 \ln x} = x^2$.

$$\begin{aligned} (x^2 y)' &= x^2(4x) = 4x^3 \Rightarrow x^2 y = x^4 + C, \\ y &= x^2 + \frac{C}{x^2}. \end{aligned}$$

Apply $y(1) = 2$: $1 + C = 2 \Rightarrow C = 1$, so $y = x^2 + \frac{1}{x^2}$.

31 Exact Equations

12. $M = 2xy$, $N = x^2 + 3y^2$. Test: $M_y = 2x = N_x$ (exact).

$$\begin{aligned} f &= \int M dx = \int 2xy dx = x^2 y + g(y), \\ f_y &= x^2 + g'(y) \stackrel{!}{=} x^2 + 3y^2 \Rightarrow g'(y) = 3y^2, \quad g = y^3. \end{aligned}$$

$$\boxed{x^2 y + y^3 = C}$$

13. $M = \cos x - y \sin x$, $N = \cos x$. Test: $M_y = -\sin x = N_x$ (exact).

$$\begin{aligned} f &= \int (\cos x - y \sin x) dx = \sin x + y \cos x + g(y), \\ f_y &= \cos x + g'(y) \stackrel{!}{=} \cos x \Rightarrow g'(y) = 0. \end{aligned}$$

$$\boxed{\sin x + y \cos x = C}$$

32 Slope Fields & Qualitative Analysis

14. Horizontal segments occur where $\frac{dy}{dx} = 0$, i.e. where the *numerator* vanishes. For $\frac{x+1}{y}$ this is $x+1=0 \Rightarrow x=-1$ — matching the given field. So the equation is

$$\frac{dy}{dx} = \frac{x+1}{y}.$$

The field is *vertical* (slope undefined) where the denominator vanishes: $y=0$.

15. $f(y) = y(2-y)$ has equilibria $y=0, 2$. For $0 < y < 2$, $f > 0$ (increasing) and for $y > 2$, $f < 0$ (decreasing):

- $y(0) = 0.5 \Rightarrow$ rises toward $y=2$.
- $y(0) = 3 \Rightarrow$ falls toward $y=2$.
- $y(0) = 2 \Rightarrow$ stays at $y \equiv 2$ (equilibrium).

Thus $y=2$ is stable, $y=0$ unstable.

33 Autonomous Equations & Phase Lines

16. $f(y) = y^2 - 4 = 0 \Rightarrow y^* = \pm 2$. With $f'(y) = 2y$:

$$f'(2) = 4 > 0 \text{ (unstable),} \quad f'(-2) = -4 < 0 \text{ (stable).}$$

Phase line: arrows point *toward* -2 and *away* from $+2$ (with $f > 0$ for $|y| > 2$, $f < 0$ for $|y| < 2$).

17. $f(P) = 0.5P(1 - P/100) = 0 \Rightarrow P^* = 0, 100$.

$$f'(P) = 0.5\left(1 - \frac{2P}{100}\right); \quad f'(0) = 0.5 > 0 \text{ (unstable),} \quad f'(100) = -0.5 < 0 \text{ (stable).}$$

Fastest growth at the vertex of f : $P=50$ (half the carrying capacity).

34 Euler's Method

18. $B_{n+1} = B_n + 0.06B_n(0.5) = B_n(1.03)$ with $\Delta t = 0.5$:

$$\begin{aligned} B_1 &= 1300(1.03) = 1339, \\ B_2 &= 1339(1.03) = 1379.17 \Rightarrow B(1) \approx 1379.17. \end{aligned}$$

19. One step $\Delta t = 1$:

$$B(1) \approx 1300 + 0.06(1300)(1) = 1378.$$

Exact: $1300e^{0.06} = 1300(1.061837) \approx 1380.39$. The two-step estimate 1379.17 is closer than the one-step 1378 — halving h reduces the error, as expected for an $O(h)$ method.

35 Existence & Uniqueness

20. $f(t, y) = y^{1/3}$ is continuous, but

$$\frac{\partial f}{\partial y} = \frac{1}{3}y^{-2/3} \text{ is discontinuous at } y=0.$$

The Picard hypothesis fails at $(0, 0)$, so uniqueness is *not* guaranteed — and indeed both

$$y \equiv 0 \quad \text{and} \quad y = \left(\frac{2}{3}t\right)^{3/2}$$

satisfy the IVP. (Existence still holds; only uniqueness is lost.)

21. Standard form: $y' + \frac{\ln t}{t-3}y = \frac{2t}{t-3}$. Coefficients p, g are continuous for $t > 0$ (because of $\ln t$) and $t \neq 3$. The initial point $t_0 = 1$ lies in the interval $(0, 3)$, so the guaranteed interval is

$$0 < t < 3$$

36 Linear Independence & the Wronskian

22.

$$W = \begin{vmatrix} e^{3t} & e^{-2t} \\ 3e^{3t} & -2e^{-2t} \end{vmatrix} = -2e^t - 3e^t = -5e^t \neq 0 \Rightarrow \text{independent.}$$

23.

$$W = \begin{vmatrix} t & 3t \\ 1 & 3 \end{vmatrix} = 3t - 3t = 0 \Rightarrow \text{dependent.}$$

Confirmed by definition: $y_2 = 3t = 3y_1$ is a scalar multiple of y_1 .

24. Standard form $y'' - \frac{3}{t}y' + \frac{4}{t^2}y = 0$, so $p = -\frac{3}{t}$. By Abel,

$$W = ce^{-\int p dt} = ce^{\int (3/t) dt} = ce^{3 \ln t} = ct^3.$$

Check with $y_1 = t^2$, $y_2 = t^2 \ln t$:

$$W = t^2(2t \ln t + t) - 2t(t^2 \ln t) = t^3. \checkmark$$

37 Second-Order Homogeneous (Three Cases)

25. $r^2 + 8r + 15 = (r+3)(r+5) = 0 \Rightarrow r = -3, -5$:

$$y = c_1 e^{-3t} + c_2 e^{-5t}.$$

26. $r^2 + 6r + 9 = (r+3)^2 = 0 \Rightarrow r = -3$ (double):

$$y = (c_1 + c_2 t) e^{-3t}.$$

27. $r^2 + r + 1 = 0$:

$$r = \frac{-1 \pm \sqrt{1-4}}{2} = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i.$$

$$y = e^{-t/2} \left(c_1 \cos \frac{\sqrt{3}}{2}t + c_2 \sin \frac{\sqrt{3}}{2}t \right).$$

28. $r^2 - 4 = 0 \Rightarrow r = \pm 2$, $y = c_1 e^{2t} + c_2 e^{-2t}$.

$$y(0) = c_1 + c_2 = 1,$$

$$y'(0) = 2c_1 - 2c_2 = 2 \Rightarrow c_1 - c_2 = 1.$$

$c_1 = 1$, $c_2 = 0$, so $y = e^{2t}$.

29. $r^2 - 1 = 0 \Rightarrow r = \pm 1$, $y = c_1 e^{-t} + c_2 e^t$.

$$y(0) = c_1 + c_2 = 2,$$

$$y'(0) = -c_1 + c_2 = 0 \Rightarrow c_1 = c_2.$$

$c_1 = c_2 = 1$, so $y = e^{-t} + e^t = 2 \cosh t$.

38 Undetermined Coefficients

30. Homogeneous $r^2 - 3r + 2 = (r - 1)(r - 2) \Rightarrow y_c = c_1 e^t + c_2 e^{2t}$. Trial $y_p = A e^{5t}$:

$$25A - 15A + 2A = 4 \Rightarrow 12A = 4 \Rightarrow A = \frac{1}{3}.$$

$$y = c_1 e^t + c_2 e^{2t} + \frac{1}{3} e^{5t}.$$

31. $y_c = c_1 \cos 2t + c_2 \sin 2t$. Trial $y_p = At^2 + Bt + C$, $y_p'' = 2A$:

$$2A + 4(At^2 + Bt + C) = 3t^2 - 1$$

$$4A t^2 + 4B t + (2A + 4C) = 3t^2 + 0t - 1.$$

Match: $4A = 3 \Rightarrow A = \frac{3}{4}$; $4B = 0 \Rightarrow B = 0$; $2A + 4C = -1 \Rightarrow C = -\frac{5}{8}$.

$$y = c_1 \cos 2t + c_2 \sin 2t + \frac{3}{4} t^2 - \frac{5}{8}.$$

32. Here $y_c = c_1 e^t + c_2 e^{2t}$ already contains e^t , so a bare $A e^t$ would be absorbed. Multiply by t :

$$y_p = A t e^t.$$

(If the overlapping root were double, multiply by t^2 .)

39 Mechanical Vibrations

33. Hooke at equilibrium $mg = kL$:

$$k = \frac{mg}{L} = \frac{(2)(9.8)}{0.5} = 39.2 \text{ N/m}.$$

$$\omega_0 = \sqrt{k/m} = \sqrt{39.2/2} = \sqrt{19.6} \approx 4.43 \text{ rad/s}, \quad T = \frac{2\pi}{\omega_0} \approx 1.42 \text{ s}.$$

34. Discriminant $\gamma^2 - 4mk = \gamma^2 - 16$:

- $\gamma = 5$: $25 - 16 = 9 > 0$ — **overdamped**.

- $\gamma = 4$: $16 - 16 = 0$ — **critically damped**.
- $\gamma = 2$: $4 - 16 = -12 < 0$ — **underdamped**, quasi-frequency

$$\mu = \frac{\sqrt{4mk - \gamma^2}}{2m} = \frac{\sqrt{12}}{2} = \sqrt{3} \approx 1.73.$$

35. $u'' + 2u' + 2u = 0 \Rightarrow r^2 + 2r + 2 = 0$:

$$r = \frac{-2 \pm \sqrt{4 - 8}}{2} = -1 \pm i.$$

$$u = e^{-t}(c_1 \cos t + c_2 \sin t).$$

Complex roots with negative real part $\Rightarrow$ **underdamped**; quasi-frequency $\mu = 1$.

End of guide. Master the purple; the black takes care of itself.